

Yicheng Su

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RESEARCH EXPERIENCE

Master's student, Julian Knight Lab, Centre of Human Genetics (Oxford)

Sep 2025 – Present

- Developed a Tiled-C differential analysis pipeline incorporating replicate-aware null modelling, distance-normalized variance estimation, and spatial smoothing, with automated stacked multi-omics (ATAC/ChIP/SNP) visualization.
- Developed a data-driven TAD detection pipeline combining insulation scores with Gaussian mixture modelling applied to HDF5 chromatin contact matrices.
- Designed and performed luciferase reporter assays to experimentally test enhancer activity of SNP-containing regulatory regions.
- Maintained reproducible workflows by repairing legacy pipelines and running nf-core / Nextflow pipelines on SLURM-based HPC clusters.
- Processed and analyzed bulk RNA-seq, ATAC-seq, and Tiled-C datasets starting from FASTQ, integrating chromatin accessibility, gene expression, and chromatin contacts to study the functional effects of disease-associated SNPs.
- Prioritized candidate regulatory variants using AlphaGenome variant effect prediction, guiding downstream CRISPR base-editing validation experiments.
- Performed ChIPmentation experiments and processed single-cell RNA-seq datasets from FASTQ to analysis-ready formats.

Research Intern, Yang Shi Lab, Ludwig Institute of Cancer Research (Oxford)

Nov 2024 - Sep 2025

- Designed and ran EC50 assays to test drug synergy for targeting AML.
- Co-developing a pan-cancer drug synergy prediction database (GUI) with Shi Lab postdoc.
- Performed bioinformatics analysis on public (DepMap and BeatAML) bulk mRNA-seq, CRISPR KO, RNAi, and drug response to identify lineage-specific vulnerabilities in AML.
- Independently identified a four-gene signature that predicts myelodysplastic syndrome (MDS) transformation to post-MDS secondary Acute Myeloid Leukemia (AML) at AUC $\approx$ 0.91 (pending for external dataset for validation) using WGCNA, mediation, and elastic-net modelling.

Model Department Intern, Biomap (Beijing)

Jul 2024 – Aug 2024

- Benchmarked protein-structure predictors; trained TorchANI model from atomic coordinates to perform machine learning force-field models calculations.

Research Intern, Hongkui Deng Lab, Peking University(Beijing)

Jul 2023 – Aug 2023

- Performed western blotting, RT-PCR, and cell passaging to confirm oxidative phosphorylation's role in stage 2 induction using human adipose-derived tissue cells into chemically induced pluripotent stem cells using the lab's small-molecule protocol published in Cell, 2023.

Research Intern, Qinghua Liu Lab, National Institute of Biological Sciences (Beijing)

Jul 2021 – Aug 2021

- Prepared reagents and assisted study of calcineurin effects on NREM sleep in mice.

EDUCATION

MBioChem Biochemistry (Molecular and Cellular), University of Oxford, Oriel College 2022 – 2026

- On track for High Upper Second-Class/First Class Honours
- Current Opinions Article: “*miRNA dosage as a regulatory principle in neurodevelopment*”

TECHNICAL SKILLS

- Bioinformatics: RNA-seq, ATAC-seq, ChIP-seq, Tiled-C processing and analysis; nf-core; working with Bash and SLURM on HPCs.
- Computational: Machine learning with Python; Data analysis using Python and R; Flutter, app development; Git; Javascript, HTML, CSS, Three.js, SQL.

INDEPENDENT PROJECTS

GuideForge - automated CRISPR gRNA design pipeline (*Independent project*)

- DOI: <https://doi.org/10.5281/zenodo.17507147>
- Inputs chromosomal coordinates and outputs ranked gRNA sequences.
- Integrates UCSC genome data, PAM scanning, quality control, and Integrated DNA technology's (IDT) on/off-target scoring.
- Allows lab-specific gRNA design rules to be configured in a policy yaml.
- Outputs a manifest per run for reproducibility.

EEG/ECG Emotion Decoding Benchmarking (*Independent Computational Neuroscience Project*)

- Signal processing using bandpower feature extraction, ICA artifact removal, windowing.
- Using Random Forest and Gradient Boosting to predict emotional states using EEG, ECG, and multimodal signals from the DREAMER dataset (DEAP validation in progress).
- Evaluating model generalizability across sample-dependent, sample-independent, and few-shot learning paradigms.

Pikkr – iOS App for Reducing Decision Fatigue (*Solo Developer*)

- Designed and launched a decision-making app on the iOS AppStore aimed at minimizing decision fatigue through probabilistic card-based selections. Implementing a multi-armed bandit algorithm to personalize decision-making.

LEADERSHIP & ACTIVITIES

Cortex Club co-president (2025-2026):

- Invited speakers and organized annual symposium
- Designed Club website with modular CMS backend (cortexclub.com)
- Recorded podcasts on Apple Podcast and Spotify
- Established partnership with China's leading neuroscience social media Neu-reality (>1M followers)

Languages: English (native); Mandarin (native); French & Spanish & Latin (intermediate)

Co-curricular Activities: Member of Oriel College Boat Club first boat (2022-2024) with 20 hours + weekly commitment; 3/78 in Torpids (Oxford's inter-college race) 2023, 1st place in Bedford Head 2023, 3rd place in National School Regatta's 15B category 2018; finished a solo 100km charity ergo under 10 hours.